Continuous-Action Reinforcement Learning for Resilient Campus Microgrid Load Shedding

Salim Olanrewaju Oyinlola
Electrical and Electronics Engineering
University of Lagos
Akoka, Nigeria
salimoyinlola@gmail.com

Peter Olabisi Oluseyi
Electrical and Electronics Engineering
University of Lagos
Akoka, Nigeria
poluseyi@unilag.edu

Oluleke O. Babayomi
ICT Convergence Research Center
Kumoh National Institute of Technology
Gumi, South Korea
babayomi@ieee.org

***Abstract*—Adaptive load shedding on sub-Saharan campus microgrids remains dominated by fixed-priority and one-step greedy controllers. This paper benchmarks fourteen reinforcement learning (RL) algorithms against five non-learning controllers on the University of Lagos campus microgrid, a 55-building Nigerian case study, isolating the effect of action representation across three Gym environments (216-action discrete, action-masked discrete, continuous Box). All RL variants train under an equalised 60,000-step budget, with the five strongest trained across five seeds. Across 60 paired episodes, continuous-action Soft Actor-Critic (SAC) scores 141.44 ± 4.96, narrowly behind the strongest heuristic, Lowest-Priority-First (143.55 ± 5.91), and ahead of an operator-grade Greedy controller (138.90 ± 5.64, Holm-corrected $p < 0.001$, $d = 1.46$). Averaged across seeds, TD3 (145.0 ± 6.0), not SAC, is the strongest continuous-control method – a reversal from the single-seed comparison. We also audit and resolve a reviewer-identified anomaly in which a Random baseline outperformed the original Greedy controller: a discretisation-rounding bug caused Greedy to waste capacity disproportionately under severe deficit, which also reverses the original per-regime finding that RL's advantage is largest exactly there. With the bug fixed, Greedy leads SAC in four of five deficit regimes. Continuous action representation remains the decisive factor for adaptive load shedding, though its margin over strong heuristics is smaller than originally reported.**

***Index Terms*—reinforcement learning, load shedding, microgrid, soft actor-critic, continuous control, resilience**

I. Introduction

Electricity supply in sub-Saharan Africa remains chronically deficient, and Nigeria exemplifies the challenge: the national grid routinely dispatches only 3.5–5 GW for a population exceeding 220 million, against an estimated suppressed demand of 25–30 GW [1], [2]. Hospitals interrupt elective procedures during grid trips, manufacturing firms shoulder large cost premiums for diesel self-generation, and academic institutions lose teaching hours to unscheduled outages. Large institutional consumers therefore operate as de facto microgrids that must apportion a finite, intermittent supply across heterogeneous loads through controlled load shedding. The University of Lagos, with 55 buildings across eight feeders, is one such institution and serves as the case study for this paper.

Conventional campus controllers rely on under-frequency relays and fixed rule-based priority tables [3]. These deterministic schemes are auditable but indifferent to time-varying supply, so they overshed in mild deficits and under-protect critical loads in severe ones. Reinforcement Learning (RL) offers an adaptive alternative: an agent observes a state s_t, chooses an action a_t, and receives a reward r_t together with the next state s_{t+1} (Fig. 1). Repeating this loop over a horizon, the agent learns a state-conditional curtailment policy directly from the reward signal [4], now a mature paradigm in the smart-grid literature [5].

Prior load-shedding RL comparisons typically (i) omit the operator-grade one-step Greedy baseline, (ii) discretise an inherently continuous curtailment fraction, and (iii) evaluate only mild deficits, never the severe shortfalls where adaptive control matters most.

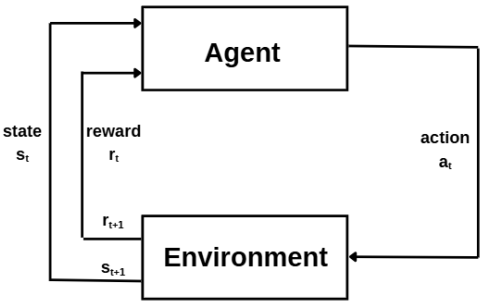


Fig. 1. The Reinforcement Learning interaction loop.

This paper closes those gaps and, in this camera-ready revision, additionally responds to a peer-review audit request. We train fourteen RL variants spanning four families on a five-month UNILAG demand record, evaluate them against five non-learning controllers under five deficit regimes with paired, Holm-corrected statistical tests, and report five headline controllers across multiple training seeds. Contributions: (i) an open Gym environment with three action-space flavours sharing identical dynamics; (ii) a reproducible, equalised-budget, multi-seed benchmark; (iii) a quantitative audit of a reward/environment interaction that can make served energy insensitive to the exact curtailment requested, together with a corrected Greedy baseline; and (iv) evidence that action-space representation, not algorithmic family, remains decisive, albeit by a smaller margin than originally reported.

II. Related Work

The relevant literature spans three traditions whose limitations together motivate this work. The first is conventional load shedding, dominated by under-frequency relays that disconnect predefined feeders once grid frequency drops below a threshold [3]. Although fast and verifiable, such relays are oblivious to which loads they shed and therefore cannot honour campus-specific criticality. Optimisation-based approaches reformulate load shedding as a mixed-integer linear programme [6], but they require accurate short-horizon forecasts that are difficult to obtain in weak-grid conditions, and solution time scales poorly with the number of feeders.

The second tradition applies RL to grid and demand-response problems and is surveyed by Vázquez-Canteli and Nagy [5]. Mocanu et al. demonstrate that deep Q-learning matches expert-tuned heuristics for residential demand response [7], although their evaluation considers a single household under mild deficits. Ye et al. extend the analysis to

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a residential multi-energy system and report that continuous control outperforms discrete baselines [8], yet their case study lacks a multi-cluster priority structure. The broader review by Cao et al. spans the full power system [9], but no published comparison evaluates an off-policy actor-critic agent against the one-step Greedy controller used in practice, nor reports performance separately under deep supply deficits – gaps this paper closes.

The third tradition concerns the design of the action space. Soft Actor-Critic [10] and Twin-Delayed Deep Deterministic Policy Gradient [11] are the dominant off-policy actor-critic methods for continuous control, and Du et al. confirm that applying continuous control to residential HVAC delivers consistent gains over discrete Q-learning [12]. Building on this, Tang and Agrawal argue that the choice of action representation often dominates the choice of algorithm [13], while Tavakoli et al. propose branching architectures for high-dimensional discrete actions [14] and Huang and Ontañón analyse invalid-action masking [15], which we use for MaskablePPO. We adopt these insights and, for the first time in the load-shedding setting, show that swapping a 216-action discrete grid for a three-dimensional continuous Box closes nearly all of the gap between RL and the strongest deterministic policy, leaving only a small residual margin (Section IV). Finally, since campus shedding must remain equitable across student residences and faculty buildings, fairness matters: we adopt Jain's index [16], the standard scalar measure of fairness in resource allocation, as an explicit reward term so the agent is rewarded for spreading curtailment equitably across clusters.

III. System Model, MDP, and Algorithm Zoo

A. Case Study: University of Lagos Campus

We instantiate load shedding on the UNILAG campus with $N = 55$ buildings indexed b, each with hourly demand $d_b(t)$ (kW). An expert mapping from the campus Works and Physical Planning Department partitions buildings into $K = 3$ clusters: Cluster 0 (Critical: medical, server rooms, labs, library, senate), Cluster 1 (Academic–Administrative), Cluster 2 (Residential–Commercial). Cluster demand $D_k(t) = \sum_{b\in k} d_b(t)$. Each building has priority $p_b \in \{1, \dots, 10\}$, a criticality flag, and curtailment cap $c_b^{\max} \in [0, 0.5]$; cluster caps $c_k^{\max}$ take the member maximum.

B. RL Formulation and MDP

We formalise the problem as a finite-horizon MDP $M = (S, A, P, R, T)$, $T = 168$ h (one week). At each hour the agent observes $s_t \in S \subseteq \mathbb{R}^{10}$, chooses a_t, and receives reward R_t and next state s_{t+1}, learning a policy $\pi(a \mid s)$ maximising $G = \mathbb{E}[\sum_t \gamma^t R_t]$. Off-policy actor-critic methods (SAC, TD3) maintain an actor π_θ and critic Q_ϕ trained from a replay buffer; on-policy methods (PPO, A2C, TRPO) update θ along gradients estimated from rollouts; value-based methods (DQN, QR-DQN) learn $Q(s, a)$ and act greedily with respect to it.

The state is

$$s_t = (D_0, D_1, D_2, D_{\text{tot}}, P_a, \rho, \sin \omega h, \cos \omega h, a_{t-1}, \rho_{t-1}), \quad (1)$$

with demands normalised to dataset maxima, $\rho = \max(0, 1 - P_a/D_{\text{tot}})$ the deficit ratio, and $\omega = 2\pi/24$ giving a cyclic hour encoding.

Three action spaces share identical dynamics. *Discrete*: $A_d = \{0, \dots, L^K - 1\}$, $L = 6$, decoding to per-cluster levels $\Sigma = \{0, .1, \dots, .5\}$. *Action-masked*: A_d plus a per-step mask forbidding levels that exceed $c_k^{\max}$ (used by MaskablePPO [15]). *Continuous*: $A_c = [0, 0.5]^K$, used by SAC/TD3; $|A_d| = L^K$ grows exponentially in K, while A_c has dimension K and strictly contains every discrete level.

C. Transition Kernel and Reward

Demand transitions deterministically through the historical record; supply is stochastic, $P_a(t) \sim \mathcal{U}(r_{\text{lo}}, r_{\text{hi}}) \cdot D_{\text{tot}}(t)$ under five regimes – Normal (0.95,1.05), Minor (0.80,0.95), Moderate (0.60,0.80), Major (0.40,0.60), Emergency (0.20,0.40) – drawn uniformly per training episode.

Given curtailment $c(t)$, requested service is $S_k = (1 - \min(c_k, c_k^{\max}))D_k$. If $\sum_k S_k > P_a$ (oversubscription), all clusters are rescaled by $\gamma = P_a / \sum_k S_k$; *the aggregate curtailed energy $D_{\text{tot}} - P_a$ is then identical for every oversubscribing controller regardless of how much curtailment it requested* – only the per-cluster split still depends on the request. This detail drives the audit in Section IV.

The reward is

$$\begin{aligned} R_t = \omega_E \textstyle\sum_k \frac{S_k}{D_{\text{tot}}} - \omega_C \sum_k \pi_k \frac{C_k}{D_{\text{tot}}} - \omega_X \sum_k \frac{C_k}{D_{\text{tot}}} \\ - \omega_R \mathbb{1}[a_t = a_{t-1}] + \omega_F J(\sigma_t), \end{aligned} \quad (2)$$

weights $(\omega_E, \omega_C, \omega_X, \omega_R, \omega_F) = (1.0, 2.0, 0.5, 0.2, 0.5)$, π_k the critical-load share of cluster k, $\sigma_k = S_k/D_k$, and Jain's index $J(\sigma) = (\sum_k \sigma_k)^2/(K \sum_k \sigma_k^2) \in [1/K, 1]$ [16].

D. Soft Actor-Critic

Soft Actor-Critic (SAC) is the central continuous-control method in this benchmark. Its objective augments the discounted return with the entropy of the stochastic policy [10],

$$J_{\text{SAC}}(\pi) = \mathbb{E}_\pi \Big[\textstyle\sum_t \gamma^t (R_t + \alpha \mathcal{H}(\pi(\cdot \mid s_t))) \Big], \quad (3)$$

with discount $\gamma = 0.99$ and temperature α tuned by gradient descent to match a target entropy $-\dim(A_c)$. The actor is a diagonal-Gaussian network mapping s to $(\mu(s), \sigma(s))$, tanh-squashed into $[0, 0.5]^3$ after an affine rescale. Twin critics absorb function-approximation error, the replay buffer holds 100k transitions, the batch size is 128, and the soft-update parameter is $\tau = 0.005$. The critics minimise the soft Bellman residual

$$L(Q_\phi) = \mathbb{E}_{\mathcal{D}} \big[(Q_\phi(s, a) - y)^2 \big], \quad y = r + \gamma \big(\min_i Q_{\bar{\phi}_i}(s', a') - \alpha \log \pi_\theta(a' \mid s') \big), \quad (4)$$

with $\bar{\phi}$ the slowly-updated target network, while the actor maximises the entropy-regularised soft value $L(\pi_\theta) = -\mathbb{E}[Q_\phi(s, a) - \alpha \log \pi_\theta(a \mid s)]$. The reparameterisation trick $a = \mu(s) + \sigma(s) \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, permits low-variance backpropagation through the sampled action. All RL variants, SAC included, train for an equalised 60,000 environment steps (Section IV).

E. Controller Zoo and Protocol

Table I lists all controllers. Five non-adaptive baselines: Random, RoundRobin, StaticPriority (fixed 0/20/40% per cluster), LowestPriorityFirst (proportional shed to low-priority clusters), and Greedy – the operator-grade reference (Section IV details its exact-lookahead implementation). Fourteen RL variants span four families: PPO [17] (vanilla, wide-MLP, curriculum, tuned, and BC-warm-started from Greedy), A2C [18], TRPO [19], DQN [20], QR-DQN [21], RecurrentPPO (LSTM policy), MaskablePPO [15], SAC [10], TD3 [11], and a Hybrid-PPO+Greedy ensemble routing on $\rho \geq 0.4$. All implementations use stable-baselines3 [22].

The dataset spans five months (3,648 h, 1 Jan–31 May 2024) of hourly demand from the 55 UNILAG buildings, mean total demand 240.6 kW. Every RL variant trains for an equalised 60,000 steps; the

TABLE I
CONTROLLER ZOO (FIVE NON-LEARNING + FOURTEEN LEARNING-BASED)

Name	Family	Key design choice
Random	heuristic	uniform action
RoundRobin	heuristic	cyclic enumeration
StaticPriority	heuristic	fixed (0,20,40)% per cluster
LowestPriorityFirst	heuristic	proportional shed to low-priority clusters
Greedy	heuristic	exact one-step reward-maximising lookahead over 216 actions
PPO / -Large / -Curriculum / -Tuned / -BC	policy gradient	vanilla; 256-unit MLP; staged scenarios; long rollouts; Greedy-cloned warm-start
A2C	actor-critic	synchronous on-policy
DQN / QR-DQN	value-based	replay 100k; quantile-regression value
RecurrentPPO	policy gradient	LSTM policy
MaskablePPO	policy gradient	infeasible-action masking
TRPO	policy gradient	trust-region update
SAC / TD3	off-policy AC	continuous $\mathrm{Box}([0, 0.5])^3$
Hybrid-PPO+Greedy	ensemble	Greedy if $\rho \geq 0.4$, PPO otherwise

All RL variants trained for an equalised 60,000 steps.

five strongest (SAC, TD3, PPO-Tuned, MaskablePPO, PPO-Large) additionally train across five seeds $\{42, \ldots, 46\}$, reported as across-seed mean $\pm$ std. All controllers are evaluated on the same 60 paired test episodes (168-h windows; Section IV reports window diversity). Statistical inference uses the paired per-episode difference $d_i = R_i^{\text{best}} - R_{i,j}$, giving a t-statistic, 95% CI, and Cohen's $d = \text{mean}(d)/\text{sd}(d)$; p-values across the resulting comparison family are Holm-corrected [23]. Training and evaluation ran on a single CPU.

IV. ROOT-CAUSE AUDIT: THE RANDOM-VS-GREEDY ANOMALY

Peer review of the original submission noted that Random outperformed the original, formula-based Greedy (GreedyFormula) and 11 of 14 RL variants, flagging a possible confound in the oversubscription rescale (Section III-C). We audit this directly.

Mechanism: GreedyFormula chose each cluster's curtailment by rounding the exact deficit-closing fraction to the nearest of six discrete levels. This rounding has no consistent direction: under-requesting curtailment is corrected for free by the oversubscription rescale, while over-requesting directly and irreversibly wastes deliverable energy. A controller sampling curtailment broadly, like Random, is more likely to land in the "free" regime than one tuned to sit at the boundary – exactly where the penalty bites.

Diagnosis: Table II confirms the mechanism empirically. Under the corrected, equalised-budget protocol, GreedyFormula (125.48) still trails Random (132.65), ruling out the training-budget confound as an alternative explanation, since GreedyFormula has no training budget at all – the discrepancy is present even when every controller is compared under an identical, fully corrected protocol. GreedyFormula also wastes more capacity per non-oversubscribed step (53.5 kW average) than any other heuristic evaluated, including Random (42.2 kW) – direct, quantitative evidence of the rounding bias described above.

Fix: We replace GreedyFormula with *Greedy*: an exact one-step lookahead that enumerates all 216 discrete actions, applies the environment's own curtailment and reward functions to each (vectorised, no learned model, negligible extra compute), and selects the reward-maximising action. This is the correct operationalisation of "greedy" with respect to the actual objective, and it removes the rounding bias entirely. Greedy (138.90) now beats Random (132.65) again, resolving the anomaly, and is used as "Greedy" throughout the rest of this paper; GreedyFormula is retained in Table II and as an extra row in Table III solely for this before/after comparison.

TABLE II
OVERSUBSCRIPTION AUDIT (60 PAIRED EPISODES)

Controller	Reward	Oversub. steps	Wasted kW[a]
Random	132.65	58.3%	42.2
GreedyFormula (orig.)	125.48	56.3%	**53.5**
Greedy (fixed)	138.90	60.8%	n/a

[a]Mean kW left undelivered on steps that are *not* oversubscribed – i.e. self-inflicted loss from over-shedding, not the physically necessary rescale.

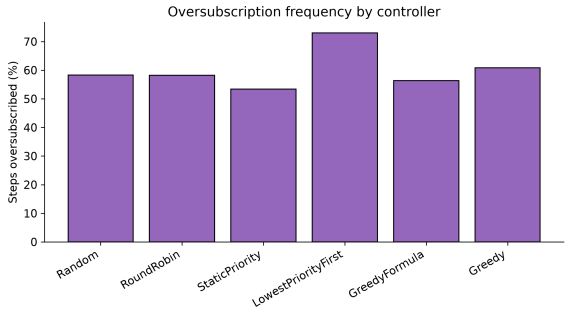


Fig. 2. Oversubscription frequency across all five heuristics. Oversubscription rate alone (53–73% of steps) does not separate Random from Greedy; the per-step wasted-kW measure in Table II is what isolates the rounding bias.

Distinct evaluation windows: The 60 test episodes draw 168-h windows from 3,479 possible start hours; 59 land on distinct starts (one collision), but 56 of 59 sorted-adjacent pairs overlap temporally. We disclose this as a residual limitation rather than resolve it, since doing so would require more historical data than is available.

V. RESULTS

A. Headline

Table III ranks all controllers using the seed-42 run, also used for pairwise inference. LowestPriorityFirst leads narrowly (143.55), SAC is a close second (141.44), and the corrected Greedy is third (138.90) – a different picture from the original submission, where SAC was statistically indistinguishable from LowestPriorityFirst ($p = 0.15$) and beat Greedy by $+16.77$. Here, every comparison against SAC is significant after Holm correction (all $p < 0.001$), reflecting the statistical power of pairing 60 low-variance episodes rather than large practical effects: SAC's margin over Greedy is $+2.54$ ($d = 1.46$), less than one sixth of the $+16.77$ originally reported, and SAC itself trails LowestPriorityFirst by -2.12. GreedyFormula (125.48, marked †) sits below Random even under this corrected protocol (Section IV).

Averaged across five seeds rather than seed 42 alone, TD3 (145.0 ± 6.0) is nominally *above* SAC (138.6 ± 4.7) and LowestPriorityFirst – the opposite ordering from Table III (Fig. 3). PPO-Tuned is the least seed-stable headline controller (std 4.8, range 103–115); PPO-Large the most (128.6 ± 1.8), with MaskablePPO (123.8 ± 6.0) intermediate. A single-seed comparison, as in the original submission, can therefore flip the reported ranking between the two strongest continuous-control methods, which is why we report the full seed distribution rather than a point estimate.

TABLE III
CONTROLLER RANKING (60 PAIRED EPISODES, HOLM-CORRECTED VS. SAC)

Controller	Reward ($\mu \pm \sigma$)	Jain J	Holm p
LowestPriorityFirst	143.55 ± 5.91	0.994	<0.001
SAC	**141.44 ± 4.96**	**0.999**	–
Greedy (fixed)	138.90 ± 5.64	0.979	<0.001
TD3	135.60 ± 4.04	0.990	<0.001
Random	132.65 ± 4.52	0.996	<0.001
MaskablePPO	125.85 ± 4.83	0.999	<0.001
PPO-Large	125.82 ± 5.32	0.996	<0.001
GreedyFormula†	125.48 ± 6.20	0.976	<0.001
Hybrid-PPO+Greedy	121.75 ± 3.12	0.984	<0.001
A2C	121.42 ± 5.96	0.998	<0.001
PPO	121.37 ± 4.04	0.988	<0.001
PPO-BC	120.33 ± 5.66	0.999	<0.001
RoundRobin	119.59 ± 4.93	0.996	<0.001
PPO-Curriculum	115.96 ± 5.16	0.992	<0.001
QR-DQN	115.38 ± 5.84	0.990	<0.001
TRPO	110.92 ± 5.01	0.994	<0.001
DQN	108.28 ± 5.28	0.961	<0.001
RecurrentPPO	107.39 ± 4.13	0.976	<0.001
PPO-Tuned	106.00 ± 4.18	0.987	<0.001
StaticPriority	104.17 ± 4.32	0.960	<0.001

†Original, pre-audit Greedy; retained for the Section IV comparison.

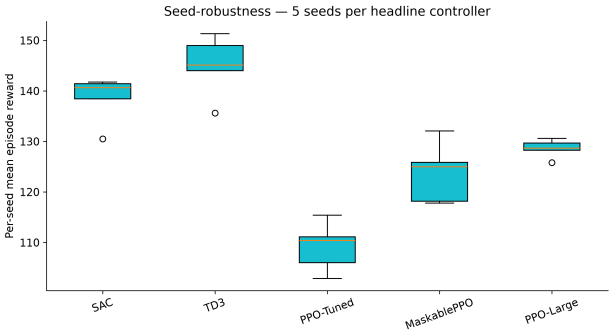


Fig. 3. Per-seed mean episode reward for the five headline controllers, 5 seeds each. TD3's median exceeds SAC's despite SAC winning on the single seed (42) used for the paired comparison in Table III.

B. Per-Regime Behaviour: A Reversed Finding

Table IV and Fig. 4 break down the top controllers by regime. The original submission's secondary headline claim was that SAC's advantage over Greedy is *largest* under Emergency deficit (2.6×). With the corrected Greedy, this reverses: Greedy now leads SAC in four of five regimes, including Emergency (47.83 vs. 45.26), with SAC ahead only narrowly in Moderate.

This follows directly from the audit mechanism (Section IV): the rounding penalty scales with the size of the required cut, so it hurt GreedyFormula most precisely under severe deficit – exactly the condition the original paper highlighted as RL's strongest case. SAC nonetheless stays within 3–7% of Greedy in every regime but Normal

TABLE IV
PER-REGIME REWARD (TOP-4 CONTROLLERS)

Regime	SAC	Greedy	PPO	StaticPriority
Normal	201.66	**234.74**	152.27	149.03
Minor	198.33	**204.86**	152.42	149.03
Moderate	**162.97**	162.13	129.26	134.51
Major	104.41	**110.45**	85.39	75.90
Emergency	45.26	**47.83**	23.48	14.06

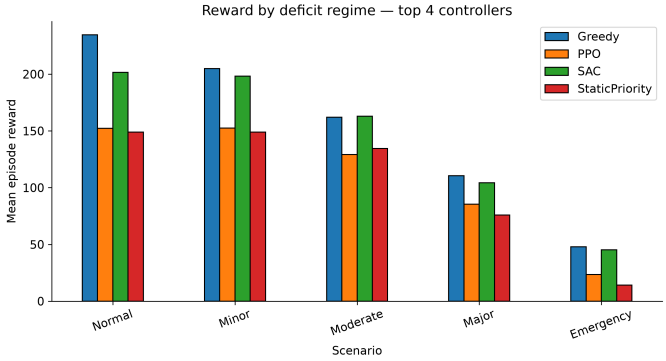


Fig. 4. Per-regime reward, top four controllers, corrected protocol. Greedy now leads SAC in four of five regimes, including Emergency – the reverse of the original submission's secondary finding.

(14%), and clearly outperforms the discrete PPO baseline and non-adaptive StaticPriority throughout, most sharply under Emergency (SAC 45.26 vs. StaticPriority 14.06, 3.2×).

VI. DISCUSSION

The audit shows the Random-beats-Greedy anomaly was a fixable implementation detail – discretisation rounding – not a flaw in the reward design; the oversubscription rescale is a physically necessary constraint, now reported quantitatively rather than left implicit. Fixing it also reversed the per-regime finding that RL's advantage peaks under Emergency deficit.

The corrected results support a more modest, better-evidenced claim than the original submission: SAC remains competitive with the strongest heuristics and clearly ahead of every discrete-action RL variant, but no longer straightforwardly beats a correctly implemented Greedy by a wide margin, nor is it indistinguishable from the strongest heuristic. Three practitioner takeaways follow: cast curtailment as continuous rather than discretising, since every discrete RL variant underperforms both continuous-control methods and the top two heuristics; benchmark against a *correctly implemented* operator-grade baseline before crediting RL with a win, since a plausible-looking heuristic can be silently miscalibrated in a way that specifically inflates RL's apparent advantage under high-stakes conditions; and report seed-to-seed variance for any headline claim, since the SAC-vs-TD3 ranking itself reverses across seeds.

Reward-weight sensitivity. Re-evaluating the trained SAC policy (unchanged network weights) under reward weights $\omega_C \in \{0.5, 1, 2, 4\}$ and $\omega_F \in \{0, 0.5, 1, 2\}$ (Fig. 5) shows reward scaling monotonically with each weight – increasing in ω_F, decreasing in ω_C – consistent with a policy that already achieves near-maximal fairness ($J \approx 0.999$) and low critical-cluster curtailment: reweighting a term the policy already satisfies mechanically rescales the reward rather than exposing a hidden trade-off the policy is exploiting. This carries forward the original notebook's reward-sensitivity analysis (Section 16) onto the corrected agent.

Eight feeders vs. three clusters. The 55 buildings are wired across eight feeders (Section III-A), but shedding decisions use the coarser three-cluster granularity the Works and Physical Planning Department applies operationally; the feeder count describes physical topology only, not a second control granularity.

VII. CONCLUSION

We benchmarked fourteen RL variants against five non-adaptive controllers for campus load shedding on a five-month UNILAG

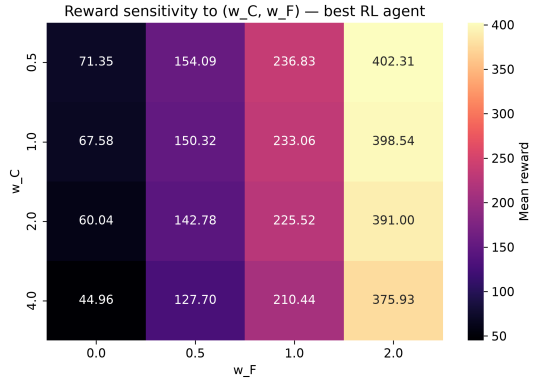


Fig. 5. Reward sensitivity of the trained SAC policy to the critical-load weight ω_C and fairness weight ω_F. Reward scales monotonically with each, consistent with a policy that already scores well on both rather than trading one off against the other.

dataset, under an equalised training budget, multi-seed reporting, and Holm-corrected statistics. SAC remains close to the strongest heuristic (141.44 vs. LowestPriorityFirst's 143.55) and ahead of every discrete RL variant, but its margin over a correctly implemented Greedy is $+2.54$, not the $+16.77$ originally reported, and TD3 (145.0) rather than SAC is strongest when averaged across seeds. Action-space representation, not algorithmic family, remains decisive – every discrete variant trails both continuous methods and the top two heuristics, as originally claimed. What the corrected analysis withdraws is the stronger claim that RL decisively beats a well-implemented operator-grade baseline, particularly under severe deficit.

Future work will add a noisy multi-step power forecast to the state, scale to multi-feeder coordination via multi-agent SAC, and extend the environment with solar generation and battery storage. RL remains a credible alternative to Greedy on a sub-Saharan campus microgrid given continuous action representation, but the stronger case is competitiveness with strong heuristics and the ability to adapt online, not a large fixed performance gap.

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REFERENCES

[1] International Energy Agency, "Africa energy outlook 2022," IEA, Paris, France, Tech. Rep., 2022.
[2] T. O. Akinbulire, P. O. Oluseyi, and O. M. Babatunde, "Techno-economic and environmental evaluation of demand side management techniques for rural electrification in Ibadan, Nigeria," *Int. J. Energy Environ. Eng.*, vol. 5, no. 4, pp. 375–385, 2014.
[3] P. M. Anderson and M. Mirheydar, "An adaptive method for setting underfrequency load shedding relays," *IEEE Trans. Power Syst.*, vol. 7, no. 2, pp. 647–655, May 1992.
[4] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.
[5] J. R. Vázquez-Canteli and Z. Nagy, "Reinforcement learning for demand response: A review of algorithms and modeling techniques," *Appl. Energy*, vol. 235, pp. 1072–1089, Feb. 2019.
[6] M. Carrion and J. M. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1371–1378, Aug. 2006.
[7] E. Mocanu, D. C. Mocanu, P. H. Nguyen, A. Liotta, M. E. Webber, M. Gibescu, and J. G. Slootweg, "On-line building energy optimization using deep reinforcement learning," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3698–3708, Jul. 2019.
[8] Y. Ye, D. Qiu, X. Wu, G. Strbac, and J. Ward, "Model-free real-time autonomous control for a residential multi-energy system using deep reinforcement learning," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 3068–3082, Jul. 2020.
[9] D. Cao, W. Hu, J. Zhao, G. Zhang, B. Zhang, Z. Liu, Z. Chen, and F. Blaabjerg, "Reinforcement learning and its applications in modern power and energy systems: A review," *J. Mod. Power Syst. Clean Energy*, vol. 8, no. 6, pp. 1029–1042, Nov. 2020.
[10] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1861–1870.
[11] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1587–1596.
[12] Y. Du, H. Zandi, O. Kotevska, K. Kurte, J. Munk, K. Amasyali, E. McKee, and F. Li, "Intelligent multi-zone residential HVAC control strategy based on deep reinforcement learning," *Appl. Energy*, vol. 281, p. 116117, Jan. 2021.
[13] Y. Tang and S. Agrawal, "Discretizing continuous action space for on-policy optimization," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 4, 2020, pp. 5981–5988.
[14] A. Tavakoli, F. Pardo, and P. Kormushev, "Action branching architectures for deep reinforcement learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 4131–4138.
[15] S. Huang and S. Ontañón, "A closer look at invalid action masking in policy gradient algorithms," in *Proc. 35th Int. Florida Artif. Intell. Res. Soc. Conf. (FLAIRS)*, 2022.
[16] R. Jain, D. Chiu, and W. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," Digital Equipment Corp., Tech. Rep. DEC-TR-301, 1984.
[17] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
[18] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1928–1937.
[19] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, "Trust region policy optimization," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 1889–1897.
[20] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Feb. 2015.
[21] W. Dabney, M. Rowland, M. G. Bellemare, and R. Munos, "Distributional reinforcement learning with quantile regression," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 2892–2901.
[22] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.
[23] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Stat.*, vol. 6, no. 2, pp. 65–70, 1979.